

Arjuna JEE 2025

Maths

DPP: 1

Permutations and Combinations

- Q1** There are 10 true-false questions in an examination. Then these questions can be answered in
(A) 240 ways (B) 20 ways
(C) 1024 ways (D) 100 ways
- Q2** A man has 3 jackets, 10 shirts and 5 pair of slacks. If an outfit consists of a jacket, a shirt, and a pair of slacks, the different outfits can the man make, is
(A) 120 (B) 150
(C) 180 (D) 210
- Q3** Number of 3-digit numbers that can be formed having unit digit as zero and repetition of digit is allowed, is
(A) 72 (B) 81
(C) 100 (D) 90
- Q4** How many numbers greater than 50000 can be formed with the digits 4, 5, 6, 7 and 8 if no digit being repeated?
(A) 96 (B) 256
(C) 218 (D) 126
- Q5** There are 6 roads between A and B and 4 roads between B and C.
(i) In how many ways can one drive from A to C by way of B ?
(ii) In how many ways can one drive from A to C and back to A, passing through B on both trips?
(iii) In how many ways can one drive the circular trip described in (ii) without using the same road more than once.
- Q6** If repetitions are not permitted, from the six digits 2, 3, 5, 6, 7 and 9 .
(i) how many 3-digit numbers can be formed
(ii) how many of these are less than 400 .
(iii) how many are even.
(iv) how many are odd.
(v) how many are multiples of 5 .
- Q7** Four dice are rolled. The number of possible outcomes in which at least one die shows 6 is
(A) 671 (B) 168
(C) 176 (D) 650
- Q8** Number of odd numbers of five distinct digits can be formed with the digits 0, 1, 2, 3, 4, is
(A) 24 (B) 120
(C) 48 (D) 36
- Q9** The number of integers greater than 6000 that can be formed, using the digits 3, 5, 6, 7 and 8 without repetition, is
(A) 120 (B) 72
(C) 216 (D) 192
- Q10** 3 different railway passes are allotted to 5 students, the number of ways this can be done is
(A) 60 (B) 20
(C) 15 (D) 10
- Q11** The number of four-digit numbers are there which are divisible by 2, is
(A) 9000 (B) 4000
(C) 4500 (D) 1000
- Q12**



The number of ways in which 5 letters be mailed, if there are 3 different mailboxes available, if each letter can be mailed in any mailbox, is

- (A) 81 (B) 243
(C) 60 (D) 125

Q13 Find the number of ways in which one can post 5 letters in 7 letter boxes.

Q14 A coin is tossed 3 times and the outcomes are recorded. How many possible outcomes are there?

Q15 How many different signals can be given using any number of flags from 4 flags of different colours?

- (A) 32 (B) 48
(C) 64 (D) 120

Q16 The number of nine non-zero digits such that all the digits in the first four places are less than the digit in the middle and all the digits in the last four places are greater than that in the middle is

- (A) 48 (B) 7560
(C) 10080 (D) 576

Q17 Find the total number of 9 digit numbers which have all the digits different

- (A) $9 \times 9!$ (B) $9!$
(C) $10!$ (D) None of these

Q18 If repetition of digits is not allowed how many numbers of four digits divisible by 5 can be formed with the digits 0, 4, 5, 6, 7

- (A) 40 (B) 44
(C) 42 (D) 36



Answer Key

Q1 (C)

Q2 (B)

Q3 (D)

Q4 (A)

Q5 (i) 24
(ii) 576
(iii) 360

Q6 (i) 120,
(ii) 40,
(iii) 40,
(iv) 80,
(v) 20

Q7 (A)

Q8 (D)

Q9 (D)

Q10 (A)

Q11 (C)

Q12 (B)

Q13 7^5

Q14 8

Q15 (C)

Q16 (D)

Q17 (A)

Q18 (C)

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